

Chapter 61

Temperature Sensor (TSENS)

61.1 Overview

ESP32-P4 provides a temperature sensor for real-time monitoring of temperature changes within the chip. The sensor converts analog voltage to digital values and provides compensation for temperature offsets.

61.2 Features

The temperature sensor has the following features:

- Software-triggered temperature measurement, which once triggered, the sensor continuously measures temperature. Software can read the data at any time.
- Hardware-triggered automatic temperature monitoring, supporting two wake-up modes
- Configurable temperature offset based on the application scenario for improved accuracy
- Configurable temperature measurement range
- Support for Event Task Matrix (ETM)-related events and tasks

61.3 Architecture

Figure [61.3-1](#) shows the internal structure of the temperature sensor.

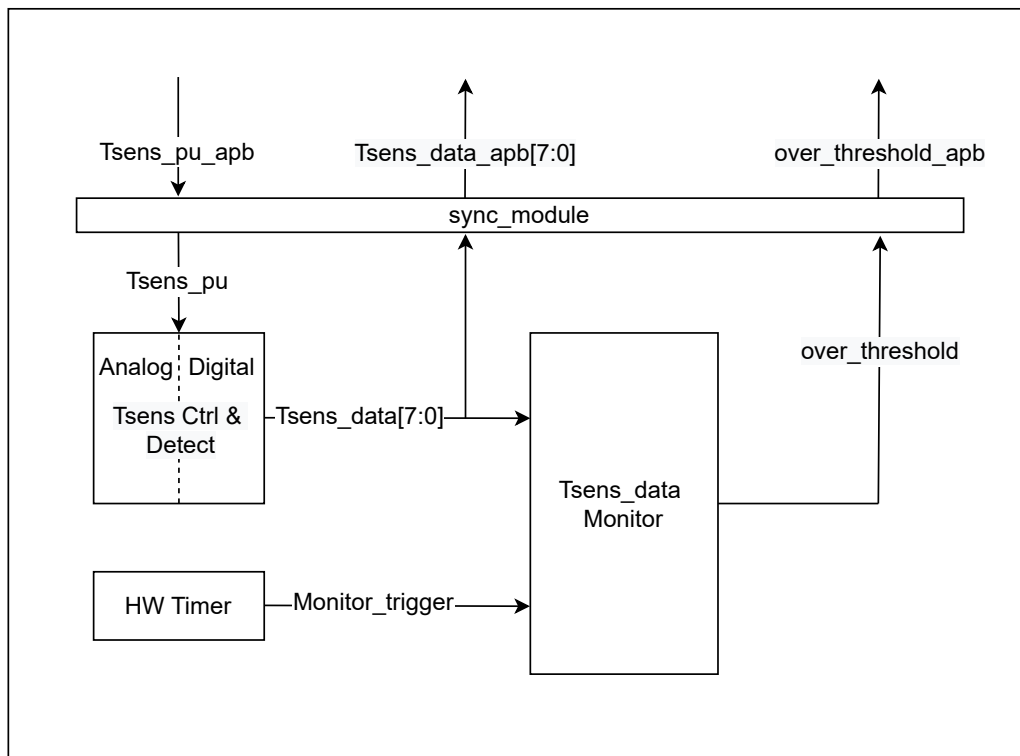


Figure 61.3-1. Temperature Sensor Architecture

As Figure 61.3-1 shows, the temperature sensor module contains the following major blocks:

- Tsens Ctrl & Detect: temperature sensor
- HW Timer: triggers automatic temperature monitoring
- Tsens_data Monitor: monitors whether the temperature is outside the threshold range
- sync_module: Synchronization block between APB clock domain and temperature sensor clock domain

61.4 Functional Description

61.4.1 Temperature Sensor Power Up

The temperature sensor can be powered up by setting the [TSENS_POWER_UP](#) register.

61.4.2 Temperature Sensor Clock

The temperature sensor has only one clock source: LP_PERI_CLK, and there is no frequency divider inside the module.

61.4.3 Wake-Up Modes for Automatic Temperature Monitoring

There are two wake-up modes for temperature monitoring, selected by [TSENS_WAKEUP_MODE](#):

- Absolute value mode:

- Monitors the absolute value of the current temperature. Configure [TSENS_WAKEUP_TH_LOW](#) and [TSENS_WAKEUP_TH_HIGH](#) to set the temperature thresholds. Wake-up will be triggered if the sampled value is above the high threshold or below the low threshold.
- Change value mode:
 - Monitors the temperature changes inside the chip. If the temperature increment of two consecutive samplings exceeds the high threshold configured in [TSENS_WAKEUP_TH_HIGH](#), or the temperature decrement of two consecutive samplings exceeds the low threshold configured in [TSENS_WAKEUP_TH_LOW](#), a wake-up will be triggered. For example, when [TSENS_WAKEUP_TH_LOW](#) is set to 8, and the two consecutive sampling values are 28 and 19, resulting in a temperature decrement of 9, a wake-up will be triggered.

61.4.4 Temperature Measurement Range and Offset

The temperature sensor is designed with five measurement ranges, as displayed in Table 61.4-1, to enhance accuracy. Each range has its own measurement errors, but users can select particular offsets to obtain calibrated data, so there is no need to be concerned about the errors.

Table 61.4-1. Temperature Measurement Range and Offset

Temperature Measurement Range (°C)	Temperature Offset
50 ~ 125	-2
20 ~ 100	-1
-10 ~ 80	0
-30 ~ 50	1
-40 ~ 20	2

61.4.5 Data Conversion

The sensor output value (*VALUE*) is stored in [TSENS_OUT](#). To calculate the actual temperature *T* (°C) based on *VALUE*, use the following equation:

$$T = 0.4386 * VALUE - 27.88 * offset - 20.52$$

where *offset* is the temperature offset displayed in Table 61.4-1.

61.5 Event Task Matrix Feature

The temperature sensor on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows the temperature sensor's ETM tasks to be triggered by any peripherals' ETM events, or temperature sensor's ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM tasks and events related to the temperature sensor. For more information, please refer to Chapter 13 [Event Task Matrix \(ETM\)](#).

The temperature sensor can receive the following ETM tasks:

- [TMPSNSR_TASK_START_SAMPLE](#): The temperature sensor starts sampling when this task is triggered.
- [TMPSNSR_TASK_STOP_SAMPLE](#): The temperature sensor stops sampling when this task is triggered.

The temperature sensor can generate the following ETM events:

- `TMPSNSR_EVT_OVER_LIMIT`: Generated when the temperature is beyond the threshold.

In practical applications, the temperature sensor's ETM events can trigger its own ETM tasks. For example, the `TMPSNSR_EVT_OVER_LIMIT` event can trigger the `TMPSNSR_TASK_STOP_SAMPLE` task.

61.6 Interrupts

ESP32-P4's Temperature Sensor can generate the following interrupt signal that will be sent to the [Interrupt Matrix](#).

- `LP_TSENS_INTR`

There is one internal interrupt source from the Temperature Sensor that can generate the above interrupt signal:

- `TSENS_COCPU_TSENS_WAKE_INT`: Triggered when the temperature sample value exceeds the threshold.

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [61.8 Register Summary](#).

61.7 Programming Procedure

The temperature sensor can be started by software as follows:

1. Set `TSENS_POWER_UP` to power up the temperature sensor.
2. Set `LPPERI_CK_EN_LP_TSENS` to enable the temperature sensor clock.
3. Wait for `TSENS_XPD_WAIT` clock cycles till the temperature sensor releases from reset and starts measuring the temperature.
4. Wait for 100 μ s (so that the output value gradually approaches the actual temperature) and then read the data from `TSENS_OUT`.

To enable hardware-triggered automatic temperature monitoring, add the following programming steps before powering up the sensor:

1. Configure `TSENS_SAMPLE_RATE` to set sampling rate.
2. Configure `TSENS_WAKEUP_MODE` to select wake-up mode for temperature monitoring.
3. Configure `TSENS_WAKEUP_TH_HIGH/LOW` to set the high and low temperature monitoring thresholds.
4. Set `TSENS_WAKEUP_EN` to start temperature monitoring.
5. Set `TSENS_SAMPLE_EN` to enable automatic temperature monitoring.

In automatic temperature monitoring mode, the output value will not be saved. Nevertheless, users can get the value at any moment by accessing `TSENS_OUT`.

61.8 Register Summary

The addresses in this section are relative to Temperature Sensor base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Control Registers			
TSENS_CTRL_REG	Temperature sensor control register 1.	0x0000	varies
TSENS_CTRL2_REG	Temperature sensor control register 2.	0x0004	R/W
Interrupt Registers			
TSENS_INT_RAW_REG	Raw register of temperature sensor interrupt.	0x0008	R/WTC/SS
TSENS_INT_ST_REG	State register of temperature sensor interrupt.	0x000C	RO
TSENS_INT_ENA_REG	Enable register of temperature sensor interrupt.	0x0010	R/WTC
TSENS_INT_CLR_REG	Clear register of temperature sensor interrupt.	0x0014	WT
TSENS_INT_ENA_W1TS_REG	TSENS_INT_ENA_REG configuration register	0x001C	WT
TSENS_INT_ENA_W1TC_REG	TSENS_INT_ENA_REG configuration register	0x0020	WT
Wake-up Control Registers			
TSENS_WAKEUP_CTRL_REG	Temperature sensor wake-up control registers.	0x0024	varies
TSENS_SAMPLE_RATE_REG	Hardware automatic sampling control registers.	0x0028	R/W

61.9 Registers

The addresses in this section are relative to Temperature Sensor base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 61.1. TSENS_CTRL_REG (0x0000)

(reserved)										TSENS_POWER_UP		TSENS_CLK_DIV		TSENS_IN_INV		(reserved)		TSENS_SAMPLE_EN		(reserved)		TSENS_OUT					
31										23		22	21		14		13	12		10		9	8	7		0	
0 0 0 0 0 0 0 0 0 0										0		0x6		0		0 0 0 0		0		0		0x0		Reset			

TSENS_OUT Stores the temperature sensor output value. (RO)

TSENS_SAMPLE_EN Configures whether to enable automatic temperature monitoring.

0: Disable

1: Enable

(R/W)

TSENS_IN_INV Configures whether to invert temperature sensor data.

0: No effect

1: Invert

(R/W)

TSENS_CLK_DIV Configures the clock division of the temperature sensor, which affects the frequency of temperature data generation by the analog circuits. (R/W)

TSENS_POWER_UP Configures whether to power up temperature sensor.

0: No effect

1: Power up

(R/W)

Register 61.2. TSENS_CTRL2_REG (0x0004)

(reserved)																TSENS_CLK_INV		TSENS_XPD_FORCE		TSENS_XPD_WAIT						
31																15		14	13	12	11				0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																1	0x0		0x2				Reset			

TSENS_XPD_WAIT Configures the wait time (in clock cycles) for the sensor to release from reset.
(R/W)

TSENS_XPD_FORCE Configures whether to enable force power up/down the temperature sensor.

- 0: Disable force power up function
- 1: Disable force power down function
- 2: Enable force power up temperature sensor
- 3: Enable force power down temperature sensor

(R/W)

TSENS_CLK_INV Configures the phase of the temperature sensor clock. (R/W)

Register 61.3. TSENS_INT_RAW_REG (0x0008)

(reserved)																															TSENS_COCPU_TSENS_WAKE_INT_RAW				
31																															1	0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

TSENS_COCPU_TSENS_WAKE_INT_RAW The raw interrupt status of the [TSENS_COCPU_TSENS_WAKE_INT](#). (R/WTC/SS)

TSENS_C0CPU_TSENS_WAKE_INT_ST

(reserved)

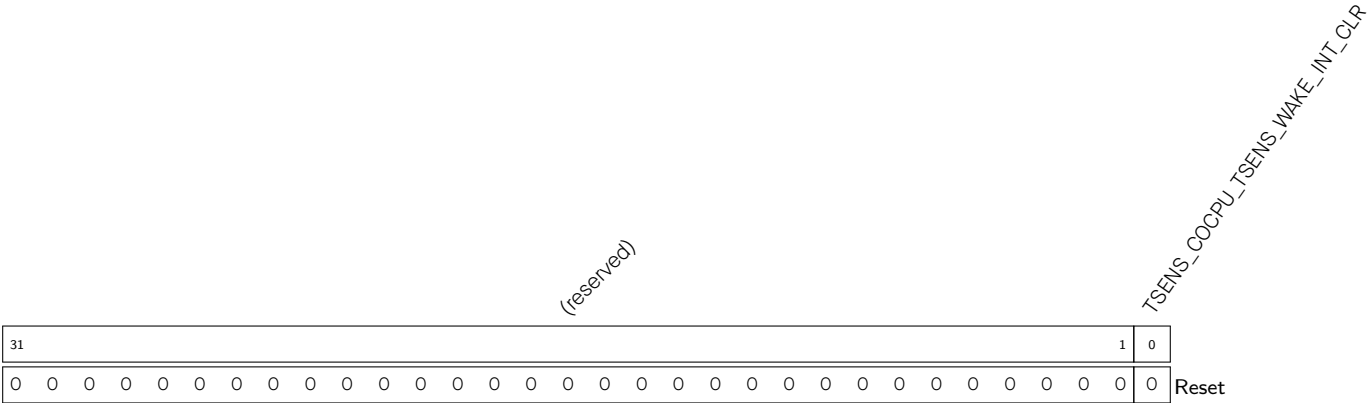
TSENS_COCPU_TSENS_WAKE_INT. (RO)

TSENS_CPU_TSENS_WAKE_INT_ENA

(reserved)

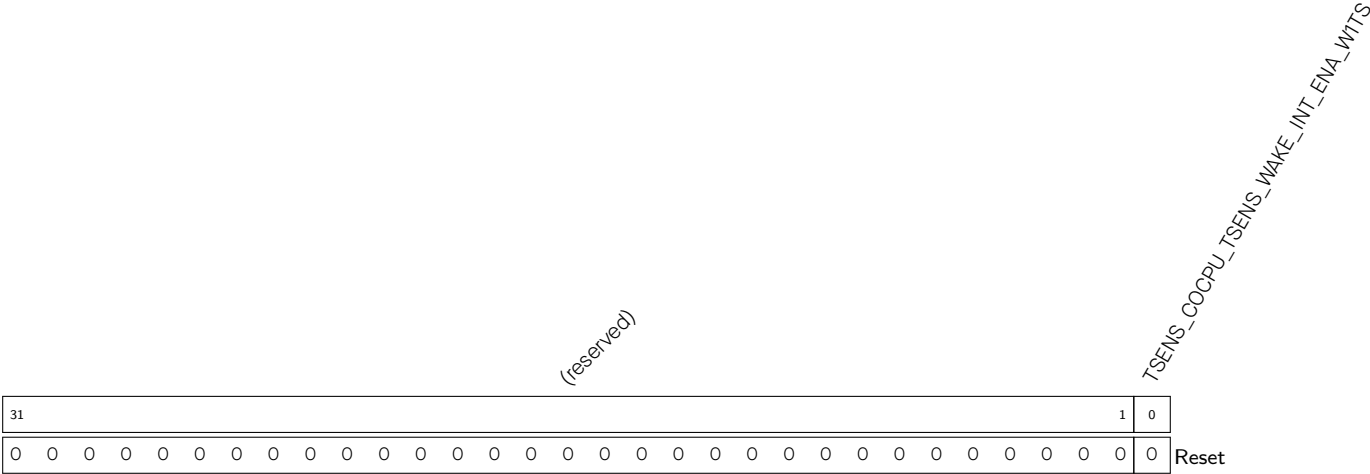
(R/WTC)

Register 61.6. TSENS_INT_CLR_REG (0x0014)



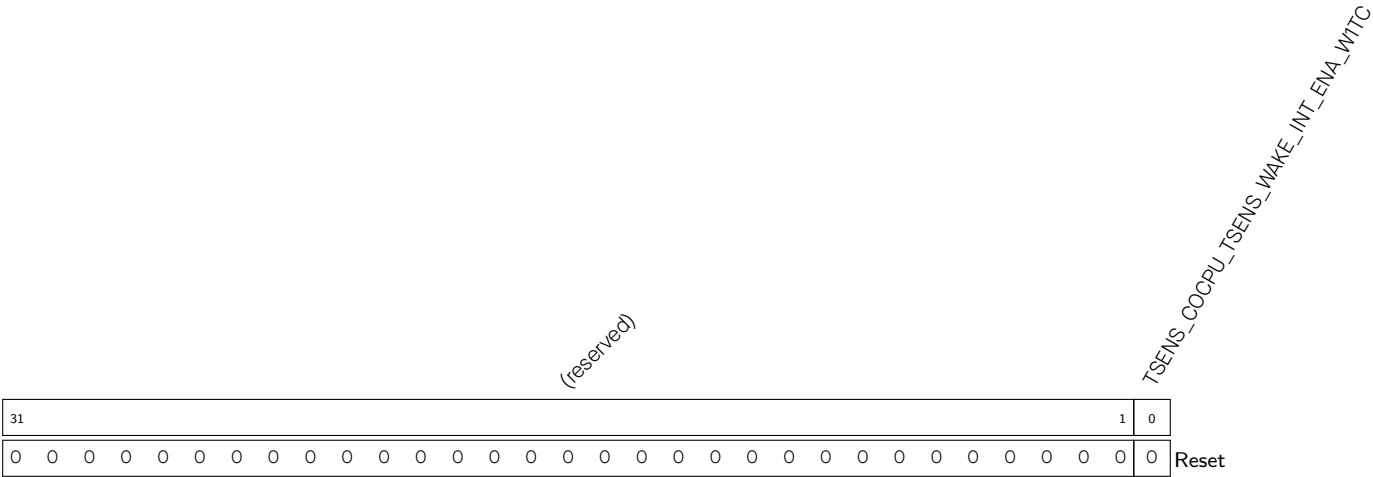
TSENS_COCPU_TSENS_WAKE_INT_CLR Write 1 to clear TSENS_COCPU_TSENS_WAKE_INT. (WT)

Register 61.7. TSENS_INT_ENA_WITS_REG (0x001C)



TSENS_COCPU_TSENS_WAKE_INT_ENA_WITS Write 1 to set TSENS_INT_ENA_REG. (WT)

Register 61.8. TSENS_INT_ENA_W1TC_REG (0x0020)



TSENS_COCPU_TSENS_WAKE_INT_ENA_W1TC Write 1 to clear TSENS_INT_ENA_REG. (WT)

Register 61.9. TSENS_WAKEUP_CTRL_REG (0x0024)

TSENS_WAKEUP_MODE																TSENS_WAKEUP_TH_HIGH																TSENS_WAKEUP_TH_LOW															
TSENS_WAKEUP_EN																(reserved)																(reserved)															
TSENS_WAKEUP_OVER_UPPER_TH																(reserved)																(reserved)															
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TSENS_WAKEUP_TH_LOW Configures the low threshold for temperature monitoring wake-up function. (R/W)

TSENS_WAKEUP_TH_HIGH Configures the high threshold for temperature monitoring wake-up function. (R/W)

TSENS_WAKEUP_OVER_UPPER_TH Represents whether the temperature output value exceeds the threshold.

0: The temperature output value is below the low threshold.

1: The temperature output value is above the high threshold.

(RO)

TSENS_WAKEUP_EN Configures whether to enable temperature monitoring wake-up function.

0: Disable

1: Enable

(R/W)

TSENS_WAKEUP_MODE Selects the wake-up mode for temperature monitoring. Valid only when [TSENS_WAKEUP_EN](#) = 1.

0: Absolute value mode

1: Change value mode

(R/W)

Register 61.10. TSENS_SAMPLE_RATE_REG (0x0028)

(reserved)																TSENS_SAMPLE_RATE	
31	16														15	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x14	

Reset

TSENS_SAMPLE_RATE Configures the sampling rate for hardware-triggered temperature monitoring. The sampling period = configured value × sensor's working clock cycle. (R/W)